

Yeri Park

ML Systems Engineer & GPU Runtime, Compiler

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Professional Summary: MSCS candidate at Georgia Tech with 4.5+ years of experience building ML system frameworks that optimize AI models on hardware accelerators, improving latency and throughput by designing hardware and software together across cloud-scale servers and embedded, on-device systems like self-driving cars and robots.

EDUCATION

Georgia Institute of Technology Atlanta, United States
Master of Science in Computer Science Sep 2025 – Present
• *Related Course:* GPU HW and SW, Advanced Operating System, HW-SW Co-Design, Deep Learning

Ewha Womans University Seoul, Republic of Korea
Bachelor of Science in Electronic and Electrical Engineering, GPA: 97.9/100 Mar 2017 – Feb 2022

EXPERIENCE

Georgia Institute of Technology Atlanta, United States
Graduate Research Intern, HPCA Lab (Advisor: Hyesoon Kim) Aug 2026 – Present
• Conduct graduate research on a RISC-V ISA-based open-source GPGPU architecture (Vortex), developing both the hardware and its compiler/runtime stack via SystemVerilog RTL, RTL simulation, and FPGA synthesis to evaluate hardware-software performance/area trade-offs
• Reduced kernel execution overhead on Vortex by profiling and re-tuning its CUDA-like compiler/driver/runtime stack (LLVM-based), pinpointing scheduling and memory-access bottlenecks across the RTL-to-software boundary

Hyundai Motor Company Seoul, Republic of Korea
Researcher, Department of Autonomous Driving SW Development Jan 2022 – Present
• Built and optimized the AI software running inside self-driving cars, making onboard computers process camera, radar, and LiDAR data faster while using less memory and power
• Reduced inference latency and memory footprint of production autonomous-driving DNNs by deploying mixed-precision quantization (PTQ/QAT), knowledge distillation, and structured pruning on NVIDIA Jetson (Thor/Orin) and Qualcomm Snapdragon; built the validation framework confirming recovered full-precision accuracy
• Improved throughput for VLM/VLA workloads via an automated optimization pipeline for layer-wise quantization and host ↔ GPU/NPU memory allocation across GPU server clusters and edge devices
• Delivered a production QNX/RTOS embedded AI inference framework (C/C++, CUDA, OpenCL) for autonomous systems, integrating car-network interfaces (LIN, CAN, Ethernet, LVDS), multimodal camera/LiDAR/Radar sensor logging, Rust-based visualization, and HILS/SILS validation across NVIDIA and Qualcomm platforms

Korea Institute of Science and Technology Seoul, Republic of Korea
Undergraduate Research Intern, Robot Vision Lab Sep 2021 – Dec 2021
• Built perception software for a wearable assistive robot, enabling it to detect walkable paths in real time using camera and LiDAR data on an embedded NVIDIA computer

Hyundai Motor Company Seoul, Republic of Korea
Undergraduate Research Intern, LiDAR Perception Development Jul 2021 – Sep 2021
• Built a real-time perception pipeline that lets a vehicle understand its surroundings from LiDAR sensor data, including detecting the ground, road edges, and nearby objects

Ewha Womans University Seoul, Republic of Korea
Undergraduate Research Intern, Digital Signal Processing Lab Jun 2019 – Sep 2019
• Developed an AI model that detects a person's breathing pattern from radar signals, for contactless health monitoring

SKILLS

Programming: C/C++, Python (PyTorch), Rust, CUDA/OpenCL, SystemVerilog/RTL, Haskell
GPU & Compiler Tooling: TensorRT, CUDA, ROCm, QNN, ONNX-MLIR, LLVM, RISC-V GPGPU (Vortex)
Model Optimization Techniques: PTQ/QAT, Knowledge Distillation, Structured/Unstructured Pruning, KV Cache Optimization
Embedded & RTOS: FreeRTOS, QNX, Linux Kernel, Yocto Project, HILS/SILS Validation
Automotive Protocols: CAN, LIN, Automotive Ethernet, LVDS
Certifications: ISTQB CTFL, ISTQB AI Testing, NVIDIA CUDA Programming
Languages: Korean (Native), English (Fluent)

ADDITIONAL PROJECTS

Edge AI Network Pruning (Aug 2025) — Applied structured/unstructured pruning to CV networks on NVIDIA Orin. C/C++, TensorRT-LLM, Swin Transformer

Rust-Based ONNX-MLIR Optimization (Apr 2025) — Built a network optimization framework for universal hardware compilation. Rust, Python, C++, ONNX-MLIR

Autonomous Driving LiDAR Perception (Dec 2023) — Ground segmentation, curb detection, free-space estimation, 3D object detection. C/C++, CUDA, OpenCL, PyTorch

HONORS & AWARDS

2nd Prize, Ewha Engineering Capstone Design Contest (2020) — AI-based audience response analysis system
Excellence Award, Korea Electric Power Engineering Camp (2020) — Top 3, national engineering camp
Academic Excellence Scholarship, Ewha Womans University — 4 consecutive semesters (2017–2021)